

- 1.** Let A be $m \times n$ and B be $n \times m$. Consider a block matrix
- 20 points

$$M = \begin{bmatrix} \mathbf{0} & A \\ -B & I_n \end{bmatrix}.$$

Find $\det M$.

Solution. Observe that

$$\begin{bmatrix} I_m & -A \\ \mathbf{0} & I_n \end{bmatrix} \begin{bmatrix} \mathbf{0} & A \\ -B & I_n \end{bmatrix} = \begin{bmatrix} AB & \mathbf{0} \\ -B & I_n \end{bmatrix}. \quad (+17 \text{ points})$$

Taking the determinant of both sides, with utilizing how determinants of block triangular matrices are computed, we get

$$\det M = \det AB. \quad (+3 \text{ points})$$

- If you split the case into $m = n$, $m > n$, and $m < n$, and then got only 1 or 2 cases correctly, then you will get at most (+7 points) or (+13 points), respectively.
- In **Section 5.1** in the lecture notes, the *Leibniz formula*

$$\det M = \sum_{\sigma : \text{permutation}} \text{sign}(\sigma) m_{1,\sigma(1)} \cdots m_{n+m,\sigma(n+m)}.$$

You can use this formula to solve this problem, but you need extra special care in computing the signs correctly. If this is not done properly, (−5 points).

- In **Section 5.2.1** in the lecture notes, the *Schur determinantal formula*

$$\det \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \det(A) \det(D - CA^{-1}B)$$

is introduced. However, this form of the Schur determinantal formula is not applicable, because in our case the upper left block is not invertible. Meanwhile, there exists another version of the Schur determinantal formula, namely

$$\det \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \det(D) \det(A - BD^{-1}C).$$

Still, this form of the Schur determinantal formula is not discussed in the lecture notes, so if you used this without any justification, (−4 points).

- 2.** Assume that a matrix B has eigenvalues 1, 2, 3, a matrix C has eigenvalues 4, 0, -4 , and a matrix D has eigenvalues -1 , -2 , -3 . Set a 6×6 matrix $A = \begin{bmatrix} B & C \\ \mathbf{0} & D \end{bmatrix}$ where B, C , and D are 3×3 matrices. We also define a linear transformation $\ell : \mathbb{R}^6 \rightarrow \mathbb{R}^6$ as $\ell(\mathbf{x}) = A\mathbf{x}$ for $\mathbf{x} \in \mathbb{R}^6$.

10+10
points

- (a) What are the eigenvalues of the 6×6 matrix A ?
- (b) Consider the unit cube Q in $\mathbb{R}^6$, that is,

$$Q = \{(x_1, \dots, x_6)^\top : 0 \leq x_i \leq 1, i = 1, \dots, 6\} \subset \mathbb{R}^6.$$

Find the volume of $\ell(Q) = \{\ell(\mathbf{x}) : \mathbf{x} \in Q\}$.

Solution. (a) Eigenvalues of A are the roots of $\det(A - \lambda I_6) = 0$. **(+3 points)** However, since $A - \lambda I_6$ is block upper triangular, $\det(A - \lambda I_6) = \det(B - \lambda I_3) \det(D - \lambda I_3)$. **(+3 points)** From this observation, eigenvalues of A are $\pm 1, \pm 2, \pm 3$. **(+4 points)**

- Correct answer without any explanation will get only **(+4 points)**.
- (b) Denote $A = [\mathbf{a}_1 | \dots | \mathbf{a}_6]$. Then $\ell(Q)$ is a parallelepiped whose edges are determined by $\ell(\mathbf{e}_i) = \mathbf{a}_i$, $i = 1, \dots, 6$. **(+4 points)** According to result from **5.3.3. Volume of a parallelograms in $\mathbb{R}^n$** in the lecture notes, volume of this parallelepiped is $|\det A|$. **(+4 points)** Since determinant of matrix is the product of all eigenvalues, we conclude that volume of $\ell(Q)$ is 36. **(+2 points)**
- Note that result of **5.3.3. Volume of a parallelograms in $\mathbb{R}^n$** is valid only when columns are linearly independent. In our case, $\det A \neq 0$ so this condition holds.
 - Correct answer without any explanation will get only **(+4 points)**.

3. Let a 4×4 matrix A is given as

10 points
each

$$A = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 2 & 0 \end{bmatrix} - 3 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 0 \end{bmatrix} - 2 \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} -1 & 1 & 0 & 0 \end{bmatrix}.$$

- Find eigenvalues and eigenvectors of A .
- Find singular values and right-/left-singular vectors of A .
- Find $\|A\|_2$ and $\|A\|_F$.
- Find the pseudoinverse A^+ of A .
- Find a 4×4 matrix B of rank 2 that minimizes $\|A - B\|_2$.

Solution. (a) First, observe that $[0 \ 0 \ 1 \ 0]^\top$, $[1 \ 1 \ 0 \ 0]^\top$, $[1 \ -1 \ 0 \ 0]^\top$ are mutually orthogonal. Let $\mathbf{v}_1 = [0 \ 0 \ 1 \ 0]^\top$, $\mathbf{v}_2 = \frac{1}{\sqrt{2}}[1 \ 1 \ 0 \ 0]^\top$, $\mathbf{v}_3 = \frac{1}{\sqrt{2}}[1 \ -1 \ 0 \ 0]^\top$. Then $\mathbf{v}_i$'s are orthonormal and

$$A = 2\mathbf{v}_1\mathbf{v}_1^\top - 6\mathbf{v}_2\mathbf{v}_2^\top + 4\mathbf{v}_3\mathbf{v}_3^\top.$$

If we multiply $\mathbf{v}_1$ both side to the right, since $\mathbf{v}_i$'s are orthonormal, we obtain $A\mathbf{v}_1 = 2\mathbf{v}_1$. By doing similar way, we can conclude that eigenpairs are $(2, \mathbf{v}_1)$, $(-6, \mathbf{v}_2)$, $(4, \mathbf{v}_3)$. Moreover, if we multiply vector that is perpendicular to $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$, for example $\mathbf{v}_4 = [0 \ 0 \ 0 \ 1]^\top$, we obtain $A\mathbf{v}_4 = \mathbf{0}$. Hence, $(0, \mathbf{v}_4)$ is also eigenpair.

- If you found 1, 2, or 3 eigenpairs, you can get **(+4 points)**, **(+6 points)**, or **(+8 points)**, respectively.
- If you at least observed that given vectors are orthogonal, you can get **(+2 points)**. This partial point will not apply if you found some of eigenpairs.
- If there is too much logical leap/lack of explanation, points will be deducted by **(-2 points)** per each logical leap/lack of explanation.

(b) We can slightly modify decomposition in (a). Let $\mathbf{u}_1 = \mathbf{v}_1$, $\mathbf{u}_2 = -\mathbf{u}_2$, $\mathbf{u}_3 = \mathbf{v}_3$. Then

$$A = 2\mathbf{u}_1\mathbf{v}_1^\top + 6\mathbf{u}_2\mathbf{v}_2^\top + 4\mathbf{u}_3\mathbf{v}_3^\top.$$

This decomposition automatically gives us a (compact) SVD :

$$A = UDV^\top, \quad U = [\mathbf{u}_2|\mathbf{u}_3|\mathbf{u}_1], \quad V = [\mathbf{v}_2|\mathbf{v}_3|\mathbf{v}_1], \quad D = \text{diag}\{6, 4, 2\}$$

and the singular pairs are

$$(2, \mathbf{u}_1, \mathbf{v}_1), (6, \mathbf{u}_2, \mathbf{v}_2), (4, \mathbf{u}_3, \mathbf{v}_3)$$

Note that $\mathbf{u}_i$'s are left singular vectors, and $\mathbf{v}_i$'s are right singular vectors.

- If you found 1 or 2 singular pairs, you can get **(+4 points)** or **(+7 points)**, respectively.
- If you didn't classify left/right singular pairs, points will be deducted by **(-2 points)**.

- (c) Recall the definition of Frobenius norm and the fact that trace is the sum of the eigenvalues, then we obtain

$$\|A\|_F = \sqrt{\text{trace}(A^\top A)} = \sqrt{\sum \sigma_i^2} = \sqrt{56}. \quad (+5 \text{ points})$$

Moreover, by **Fact 7.1**, $\|A\|_2 = \sigma_1 = 6$. (+5 points)

- If you exactly know the methods but your answer is wrong due to the (b), you can receive (+5 points).
- (d) From compact SVD we obtained in (b),

$$A^\dagger = VD^\dagger U^\top = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1/6 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1/2 \end{bmatrix} \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

- Decomposed form is okay.
 - If you exactly know the methods but your answer is wrong due to the (b), you can receive (+5 points).
 - If you didn't sorted singular values by their magnitude, points will be deducted by (-2 points).
- (e) By **Fact 8.1**,

$$B = A_2 = 6\mathbf{u}_2\mathbf{v}_2^\top + 4\mathbf{u}_3\mathbf{v}_3^\top$$

- Decomposed form is okay.
- If you exactly know the methods but your answer is wrong due to the (b), you can receive (+5 points).

4. Suppose A is a 3×4 matrix given as $A = \begin{bmatrix} 2 & 1 & -1 & 1 \\ 4 & 2 & -2 & 2 \\ -2 & -1 & 1 & -1 \end{bmatrix}$.

10 points

each

- Find eigenvalues and eigenvectors of $A^\top A$.
- Find eigenvalues and eigenvectors of $AA^\top$.
- Is there a 3×4 matrix B of rank 2 that minimizes $\|A - B\|_2$?

Solution. (a) Note that A has rank 1.

We can write $A = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 & 1 \end{bmatrix} = \sqrt{42} \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{-1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} \frac{2}{\sqrt{7}} & \frac{1}{\sqrt{7}} & \frac{-1}{\sqrt{7}} & \frac{2}{\sqrt{7}} \end{bmatrix}$.

Let $\mathbf{u} = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{-1}{\sqrt{6}} \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} \frac{2}{\sqrt{7}} \\ \frac{1}{\sqrt{7}} \\ \frac{-1}{\sqrt{7}} \\ \frac{2}{\sqrt{7}} \end{bmatrix}$, then $A = \sqrt{42}\mathbf{u}\mathbf{v}^\top$ is a SVD of A by **Fact 7.2**.

So 42 is the only nonzero eigenvalue of $A^\top A$ and $AA^\top$, hence the eigenvalues of $A^\top A$ are 0 and 42. Moreover, $\mathbf{v}$ is the eigenvector of $A^\top A$ associated to 42. Finally, by computing the nullspace of $A^\top A$, one can conclude that $\left\{ \begin{bmatrix} -\frac{1}{2} & 1 & 0 & 0 \end{bmatrix}^\top, \begin{bmatrix} \frac{1}{2} & 0 & 1 & 0 \end{bmatrix}^\top, \begin{bmatrix} -\frac{1}{2} & 0 & 0 & 1 \end{bmatrix}^\top \right\}$ are eigenvectors of $A^\top A$ associated to 0.

- As $A^\top A$ and $AA^\top$ share nonzero eigenvalues, $AA^\top$ has 0 and 42 as its eigenvalues. By the SVD of A given in (a), we see that $\mathbf{u}$ is the eigenvector of $AA^\top$ associated to 42. Again by finding the nullspace, we have $\left\{ \begin{bmatrix} -2 & 1 & 0 \end{bmatrix}^\top, \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^\top \right\}$ as the eigenvectors of $AA^\top$ associated to 0.
- No. First, it is clear that there is no 3×4 matrix B with rank 2 such that $\|A - B\|_2 = 0$, because A has rank 1. Now let $\|A - B\|_2 = \epsilon > 0$. Let $M := A - \frac{1}{2}(A - B)$, then clearly M is a 3×4 matrix of rank 2. However, $\|A - M\|_2 = \|\frac{1}{2}(A - B)\|_2 = \frac{\epsilon}{2} < \epsilon$. So we can always get another rank 2 matrix with smaller $\|A - B\|_2$. Hence, there is no 3×4 matrix of rank 2 that minimizes $\|A - B\|_2$.

- Each problem is worth **(+10 points)**.
- In (a) and (b), if you only found the eigenpairs corresponding to the nonzero eigenvalues, you can only get **(+5 points)** each.
- In (a) and (b), if you found the correct eigenvalues but incorrect eigenvectors or vice versa, you were deducted **(-2 points)** for each case.
- For (c), if you claimed that there is no such matrix B but failed to give any justification, you only get **(+2 points)**. Moreover, if you only wrote that there is no such B because A has rank 1, you cannot get more than **(+5 points)**.

5. Let A be an $m \times n$ matrix. Define its symmetrization $s(A)$ as

10 points
each

$$s(A) = \begin{bmatrix} \mathbf{0} & A \\ A^\top & \mathbf{0} \end{bmatrix}$$

which is an $(m+n) \times (m+n)$ **symmetric** matrix as the name suggests.

- Let $(\sigma, \mathbf{v}, \mathbf{u})$ be a singular triplet of A . Then, σ and $-\sigma$ are eigenvalues of $s(A)$. Find the eigenvectors of $s(A)$ associated with the eigenvalues σ and $-\sigma$, respectively.
- Let $(\lambda, \mathbf{w})$ be an eigenpair of $s(A)$ where $\lambda \neq 0$. Then, $-\lambda$ is also an eigenvalue of $s(A)$. Find the eigenvector of $s(A)$ associated with the eigenvalue $-\lambda$.
- Let $\lambda < 0$ and $(\lambda, \mathbf{w})$ be an eigenpair of $s(A)$. Find right-/left-singular vectors of A associated with the singular value $-\lambda$.

Solution. (a) Since $(\sigma, \mathbf{v}, \mathbf{u})$ is a singular triplet of A , we have $A\mathbf{v} = \sigma\mathbf{u}$, and $A^\top\mathbf{u} = \sigma\mathbf{v}$. Note that $\mathbf{u}$ is an $m \times 1$ matrix while $\mathbf{v}$ is an $n \times 1$ matrix. Let $\mathbf{w}$ be an $(m+n) \times 1$ matrix such that $\mathbf{w}^\top = [\mathbf{u}^\top \ \mathbf{v}^\top]$. Then $s(A)\mathbf{w} = \begin{bmatrix} 0 & A \\ A^\top & 0 \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ \mathbf{v} \end{bmatrix} = \begin{bmatrix} A\mathbf{v} \\ A^\top\mathbf{u} \end{bmatrix} = \begin{bmatrix} \sigma\mathbf{u} \\ \sigma\mathbf{v} \end{bmatrix} = \sigma \begin{bmatrix} \mathbf{u} \\ \mathbf{v} \end{bmatrix} = \sigma\mathbf{w}$. Thus, $\mathbf{w}$ is an eigenvector of $s(A)$ associated with the eigenvalue σ .

Similarly, let $\mathbf{w}' = [\mathbf{u}^\top \ (-\mathbf{v})^\top]^\top$, then we have $s(A)\mathbf{w}' = (-\sigma)\mathbf{w}'$. Thus, $\mathbf{w}'$ is an eigenvector of $s(A)$ associated with the eigenvalue $-\sigma$.

- Write $\mathbf{w} = (w_1, \dots, w_{m+n})^\top$, and let $\mathbf{w}' = (w_1, \dots, w_m, -w_{m+1}, \dots, -w_{m+n})^\top$. For simplicity, write $\mathbf{w}_1 = (w_1, \dots, w_m)^\top$, $\mathbf{w}_2 = (w_{m+1}, \dots, w_{m+n})^\top$. Then $\mathbf{w}' = [\mathbf{w}_1^\top \ -\mathbf{w}_2^\top]^\top$ and we have $s(A)\mathbf{w}' = [(-A\mathbf{w}_2)^\top \ (A^\top\mathbf{w}_1)^\top]^\top = [-\lambda\mathbf{w}_1 \ \lambda\mathbf{w}_2]^\top = (-\lambda)\mathbf{w}'$. Thus, $\mathbf{w}'$ is the eigenvector of $s(A)$ associated with the eigenvalue $-\lambda$.

- As $(\lambda, \mathbf{w})$ is an eigenpair of $s(A)$, we have $s(A)\mathbf{w} = \lambda\mathbf{w}$. Write $\mathbf{w} = [\mathbf{w}_1^\top \ \mathbf{w}_2^\top]^\top$, where $\mathbf{w}_1 = (w_1, \dots, w_m)^\top$, $\mathbf{w}_2 = (w_{m+1}, \dots, w_{m+n})^\top$. Then

$$(s(A))^\top s(A)\mathbf{w} = s(A)^\top \lambda\mathbf{w} = \lambda(s(A)\mathbf{w}) = \lambda^2\mathbf{w}.$$

In other words, $\mathbf{w}$ is an eigenvector of $s(A)^\top s(A) = s(A)s(A)^\top$ associated to the eigenvalue λ^2 . Since $s(A)$ is symmetric, $\mathbf{w}$ is an eigenvector of $(s(A))^2$ associated to λ^2 . Thus we have

$$\begin{bmatrix} 0 & A \\ A^\top & 0 \end{bmatrix} \begin{bmatrix} 0 & A \\ A^\top & 0 \end{bmatrix} \begin{bmatrix} \mathbf{w}_1 \\ \mathbf{w}_2 \end{bmatrix} = \begin{bmatrix} AA^\top & 0 \\ 0 & A^\top A \end{bmatrix} \begin{bmatrix} \mathbf{w}_1 \\ \mathbf{w}_2 \end{bmatrix} = \begin{bmatrix} \lambda^2\mathbf{w}_1 \\ \lambda^2\mathbf{w}_2 \end{bmatrix}.$$

Hence we have $AA^\top\mathbf{w}_1 = \lambda^2\mathbf{w}_1$ and $A^\top A\mathbf{w}_2 = \lambda^2\mathbf{w}_2$. Let $\mathbf{w}'_1 := \frac{\mathbf{w}_1}{\|\mathbf{w}_1\|}$, $\mathbf{w}'_2 := \frac{\mathbf{w}_2}{\|\mathbf{w}_2\|}$. Then by (a) and (b), $-\mathbf{w}'_2$ and $\mathbf{w}'_1$ are the left and right singular vectors of A associated with the singular value $-\lambda$, respectively. Therefore, $(-\lambda, -\mathbf{w}'_2, \mathbf{w}'_1)$ is the desired singular triplet of A .

- Even if you have a different presentation for the eigenvectors, you can still get full credit. For instance, choosing $\mathbf{w}' = [(-\mathbf{u})^\top \ \mathbf{v}^\top]^\top$ in (a) is a valid answer.
- In (c), if you did not normalize the eigenvector to get the singular vectors, you were deducted **(-5 points)**.

6. Let A be an $m \times n$ matrix with at least one non-zero entry. What are the projection
 5 points $\mathbf{P}_{\text{Col}(A)}$ onto $\text{Col}(A)$ and the pseudoinverse A^+ of A if
 each

- (a) A consists of a single column $\mathbf{v}$, that is, $n = 1$;
- (b) the columns of A are orthonormal;
- (c) the columns of A are linearly independent (that is $\text{rank } A = n$);
- (d) the columns of A are possibly linearly dependent and one of its compact SVD is given as $U\Sigma V^\top$.

Solution. (a) By the assumption that A has at least one non-zero entry, $\mathbf{v} \neq \mathbf{0}$. Clearly, A is of rank 1, thus it is immediate that $A = \frac{\mathbf{v}}{\|\mathbf{v}\|} \begin{bmatrix} \|\mathbf{v}\| \end{bmatrix} \begin{bmatrix} 1 \end{bmatrix}^\top$ is a compact SVD. Therefore, we get

$$A^+ = \frac{\mathbf{v}^\top}{\|\mathbf{v}\|^2}, \quad (+5 \text{ points})$$

$$\mathbf{P}_{\text{Col}(A)} = AA^+ = \frac{1}{\|\mathbf{v}\|^2} \mathbf{v}\mathbf{v}^\top.$$

(b) As the columns of A are orthonormal, $A = AI_n I_n$ is in fact a compact SVD of A . Hence,

$$A^+ = I_n^\top I_n^{-1} A^\top = A^\top, \quad (+5 \text{ points})$$

$$\mathbf{P}_{\text{Col}(A)} = AA^+ = AA^\top.$$

(c) In this case, $A^\top A$ is invertible, so we can use the formulae

$$A^+ = (A^\top A)^{-1} A^\top, \quad (+5 \text{ points})$$

$$\mathbf{P}_{\text{Col}(A)} = AA^+ = A(A^\top A)^{-1} A^\top.$$

(d) By definition we have

$$A^+ = V\Sigma^{-1}U^\top, \quad (+2 \text{ points})$$

and because $V^\top V = I$ it follows that

$$\begin{aligned} \mathbf{P}_{\text{Col}(A)} &= AA^+ = U\Sigma V^\top V\Sigma^{-1}U^\top \\ &= U\Sigma\Sigma^{-1}U^\top \\ &= UU^\top. \end{aligned} \quad (+3 \text{ points})$$

- For (a), if you did not normalize $\mathbf{v}$, (**-5 points**).
- For (a) and (b), if you simply wrote $A^+ = (A^\top A)^{-1} A^\top$ as the final answer, (**-4 points**).
- For (a), (b), and (c), if you only got either A^+ or $\mathbf{P}_{\text{Col}(A)}$ correct, (**+3 points**).
- For (a), (b), (c), and (d), if you did not specify whether your answer is for A^+ or for $\mathbf{P}_{\text{Col}(A)}$, (**-2 points**).
- No deduction of points beyond (**0 points**).

7. Let A be an $n \times n$ symmetric positive definite matrix. Denote its eigenvalues as $\lambda_1, \dots, \lambda_n$. Let B be the square root of A , that is, $A = B^2$.

5 points

each

- Let $A = U^\top U$ be the Cholesky factorization of A . Find $|\det U|$.
- Find the eigenvalues of B .
- Let $B = QR$ be the QR -decomposition of B . Find $|\det R|$.
- Does there exist the square root C of B ? If yes, what are the eigenvalues of C ?

Solution. (a) Because A is positive definite, $\det A = \lambda_1 \cdots \lambda_n > 0$. So, from

$$\det A = \det(U^\top U) = \det U^\top \det U = (\det U)^2 \quad (+4 \text{ points})$$

we have $|\det U| = \sqrt{\det A} = \sqrt{\lambda_1 \cdots \lambda_n}$. (+1 point)

- (b) Let $A = P^\top \Lambda P$ be the spectral decomposition with orthogonal P and $\Lambda = \text{diag}\{\lambda_1, \dots, \lambda_n\}$. Then by definition, $B = P^\top \Lambda^{1/2} P$ where $\Lambda^{1/2} = \text{diag}\{\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}\}$. But this representation of B is already in the form of a spectral decomposition, so the eigenvalues of B are $\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}$. (+5 points)

- (c) Because Q is orthogonal, $1 = \det I_n = \det(Q^\top Q) = (\det Q)^2$, so $|\det Q| = 1$. Therefore, from

$$|\det B| = |\det(QR)| = |\det Q| |\det R| = |\det R| \quad (+4 \text{ points})$$

we get $|\det R| = |\det B| = \sqrt{\lambda_1 \cdots \lambda_n}$. (+1 point)

- (d) Being a square root of a positive definite matrix, B itself is also positive definite. Thus, there does exist a square root C of B . (+3 points) Indeed, defining $\Lambda^{1/4} = \text{diag}\{\sqrt[4]{\lambda_1}, \dots, \sqrt[4]{\lambda_n}\}$ so that $(\Lambda^{1/4})^2 = \Lambda^{1/2}$, it is clear that $C = P^\top \Lambda^{1/4} P$ is a square root of B .

Moreover, $C = P^\top \Lambda^{1/4} P$ is already in the form of a spectral decomposition, so the eigenvalues of C are $\sqrt[4]{\lambda_1}, \dots, \sqrt[4]{\lambda_n}$. (+2 points)

- For each minor mistake, (−1 point).
- If the answers for (a) and (c) are both wrong but if that is because you made a mistake in computing $\sqrt{\det A}$, then (−1 point) in total.
- For (d), if you correctly computed the eigenvalues of C but did not show why C exists, you can get only at most (+3 points).